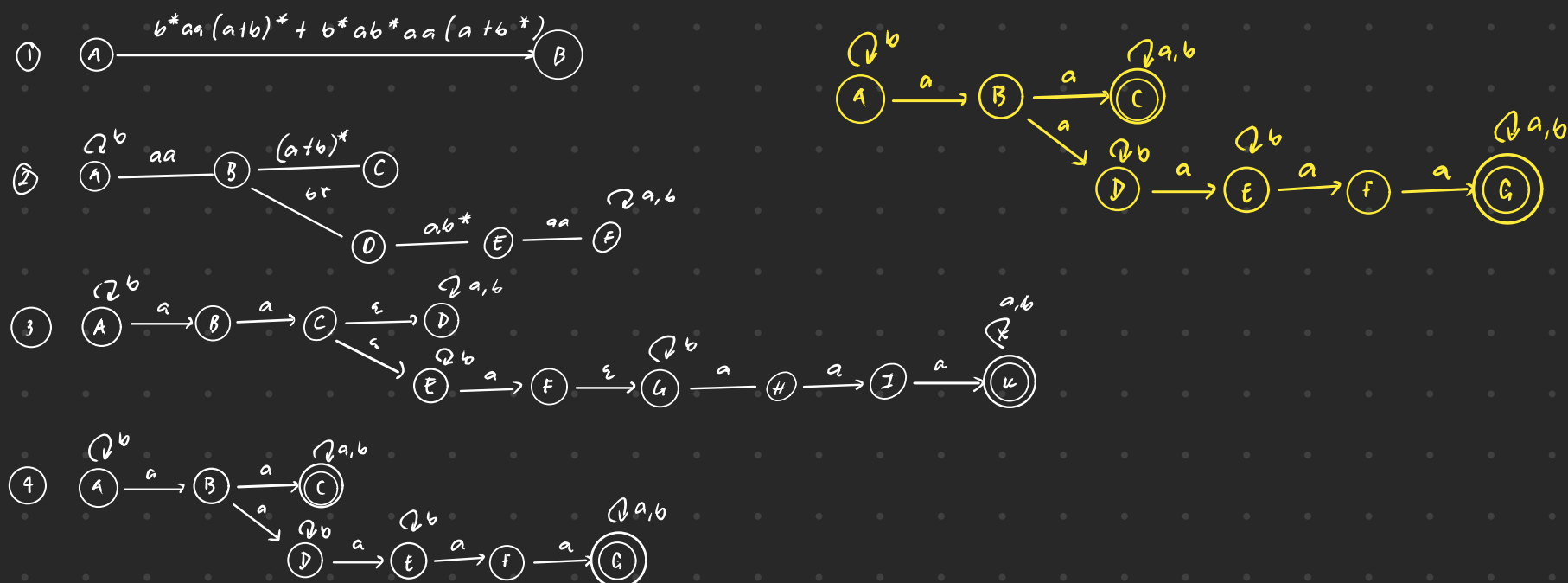


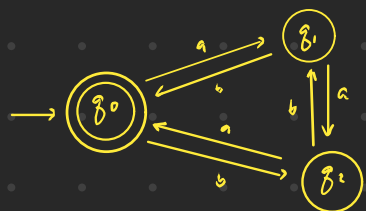
1. Regular Expression to Finite Automata

$$R = b^*aa(ab)^* + b^*ab^*aa(a+b^*)$$



Source used in solving this problem: [Tutorial's Point](#)

2. Finite Automata to Regular Expression



Initial equations

$$q_0 = \epsilon + q_1 b + q_2 a \quad (1)$$

$$q_1 = q_0 a + q_1 a \quad (2)$$

$$q_2 = q_1 a + q_0 b \quad (3)$$

1. Substituting q_1 and q_2 into the equation of q_0

$$\begin{aligned} q_0 &= \epsilon + q_1 b + q_2 a \\ &= \epsilon + (q_0 a + q_1 a) b + (q_1 a + q_0 b) a \\ &= \epsilon + q_0 ab + q_1 ab + q_1 aa + q_0 ba \end{aligned}$$

2. Substitute $q_1 = q_0 a + q_2 b$

$$\begin{aligned} &= \epsilon + q_0 ab + q_1 ab + q_1 aa + q_0 ba \\ &= \epsilon + q_0 ab + q_1 ab + (q_0 a + q_2 b) aa + q_0 ba \\ &= \epsilon + q_0 ab + q_1 ab + q_0 aaa + q_2 baa + q_0 ba \\ &= \epsilon + q_0 (ab + aaa + ba) + q_1 (ab + baa) + q_2 (ba + baa) \end{aligned}$$

3. Substitute $q_2 = q_1 a + q_0 b$

$$\begin{aligned} &= \epsilon + q_0 (ab + aaa + ba) + q_1 (ab + baa) + q_2 (ba + baa) \\ &= \epsilon + q_0 (ab + aaa + ba) + (q_1 a + q_0 b) (ba + baa) \\ &= \epsilon + q_0 (ab + aaa + ba) + q_1 aba + q_0 bba + q_0 baa \end{aligned}$$

4. Substituting $q_1 = q_0 a + q_2 b$ again

$$\begin{aligned} &= \epsilon + q_0 (ab + aaa + ba) + q_1 aba + q_0 bba + q_0 baa \\ &= \epsilon + q_0 (ab + aaa + ba) + (q_0 a + q_2 b) aba + q_0 bba + q_0 baa \\ &= \epsilon + q_0 (ab + aaa + ba) + q_0 aaa + q_2 bba + q_0 bba + q_0 baa \end{aligned}$$

5. Substituting $q_2 = q_1 a + q_0 b$ one last time

$$\begin{aligned} &= \epsilon + q_0 (ab + aaa + ba) + q_0 aaa + q_2 bba + q_0 bba + q_0 baa \\ &= \epsilon + q_0 (ab + aaa + ba) + q_0 aaa + (q_1 a + q_0 b) bba + q_0 bba + q_0 baa \\ &= \epsilon + q_0 (ab + aaa + ba) + q_0 aaa + q_1 abba + q_0 bba + q_0 bba + q_0 baa \\ &= \epsilon + q_0 (ab + aaa + ba) + q_0 aaa + q_0 abba + q_0 bba + q_0 bba + q_0 baa \end{aligned}$$

$$= \epsilon + q_0(ab + aqa + ba) + q_0 aa(bb + baq) + q_0 aaba(bb + baq) + q_1 baba(bb + baq) + q_0 bba + q_0 b(bb + baq)$$

6 Isolating q_0

$$= \epsilon + q_0(ab + aqa + ba + aa(bb + baq) + aaba(bb + baq) + bba(bb + baq) + b(bb + baq)) + q_1 baba(bb + baq)$$

7 Applying Arden's Theorem

$$q_0 = (\epsilon + q_1 baba(bb + baq))^+ (ab + aqa + ba + aa(bb + baq) + aaba(bb + baq) + bba(bb + baq) + b(bb + baq))$$

8. The regular expression of the given DFA is

$$q_0 = (\epsilon + q_1 baba(bb + baq))^+ (ab + aqa + ba + aa(bb + baq) + aaba(bb + baq) + bba(bb + baq) + b(bb + baq))$$

Source used in solving this problem

Neso Academy DFA to Regular Expression